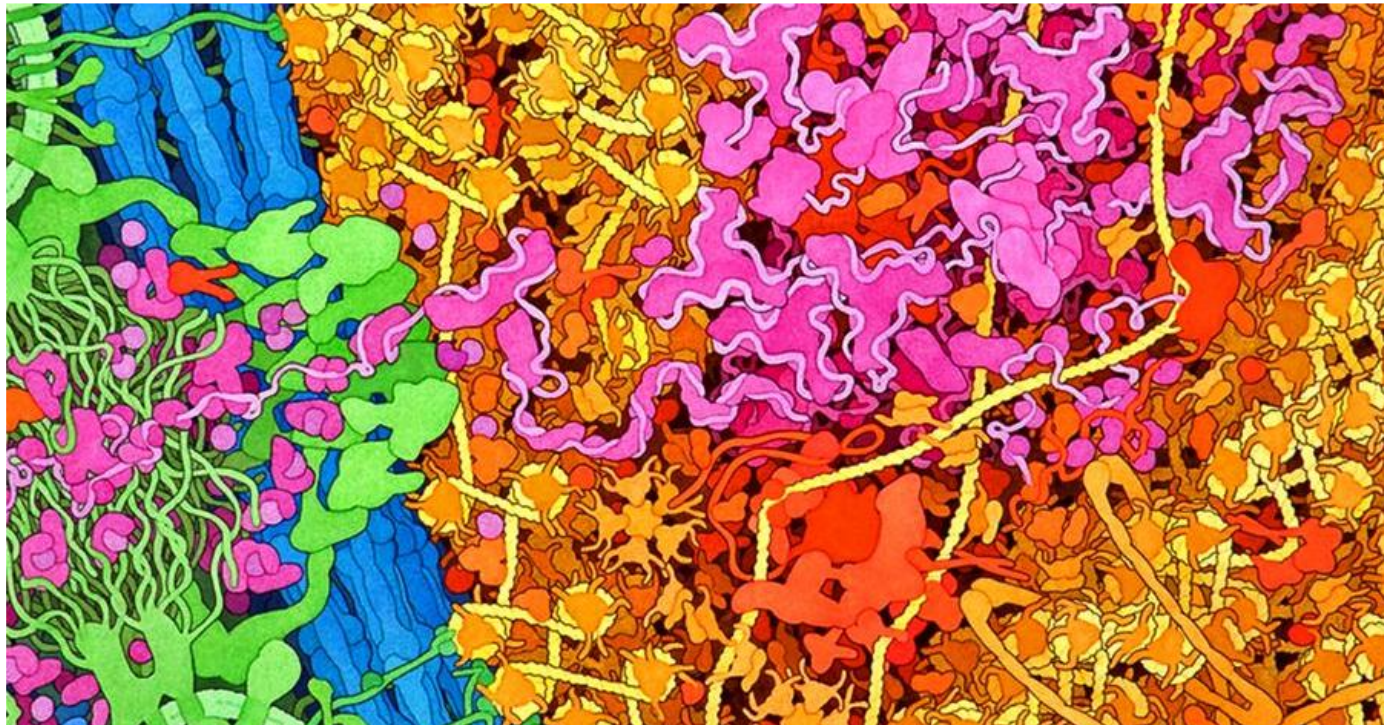


Chapter 5

How to reconcile the tendency to increase disorder with the highly controlled and structured processes in life?



$$S(N, V, E) = k_B [\ln (A_{3N} (2mE)^{3N/2}) + \ln V^N - \ln N! - \ln (2\pi\hbar)^{3N}] \quad (5.5)$$

Image credits: David Goodsell

excluded volume interactions

Entropy effect

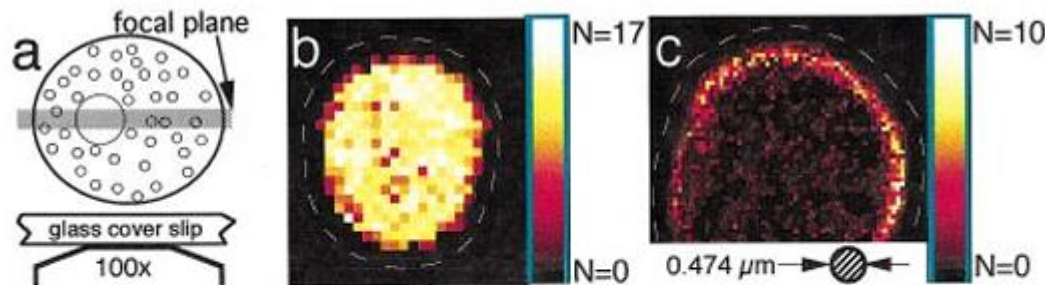
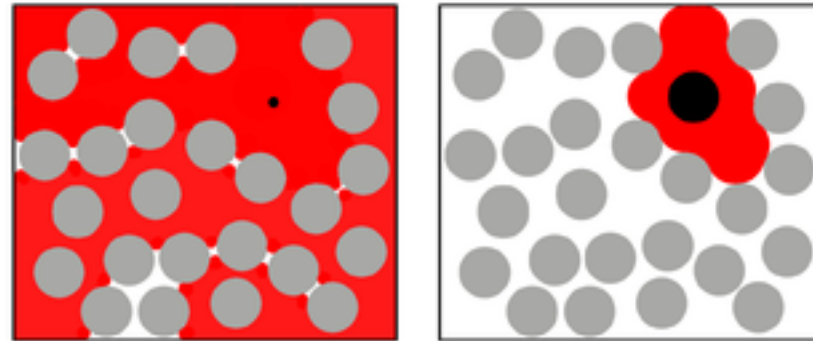
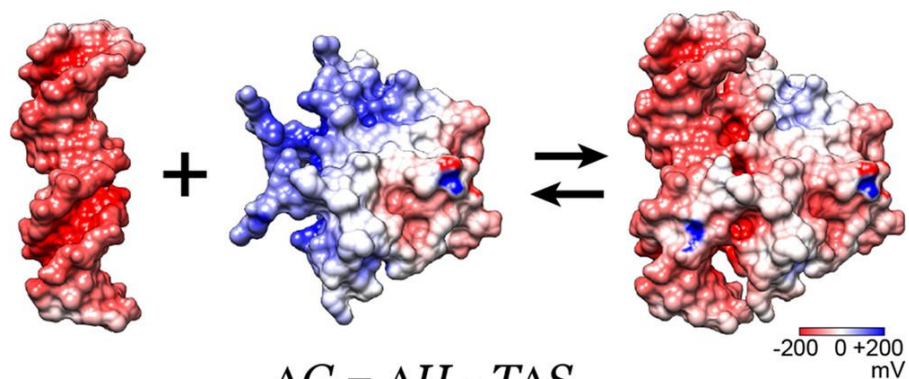


FIG. 1. (a) Cartoon of the 600-nm thick slice through an SOPC vesicle imaged with an optical microscope. We measured the in-plane positions of the larger colloidal sphere when it was in focus. (b)(color) Probability distribution of a single 0.237- μm -radius polystyrene sphere inside a vesicle (no small spheres). The white dashed line is the edge of the vesicle, and the colored points indicate the number of times, N , the center of the sphere was observed in a bin located at a given point. There were 2000 events and the bins were 130×130 nm. The sphere simply diffused freely throughout all of the available space. (c)(color) Same as in (b), but with a vesicle that also contained small spheres ($\phi_s = 0.30$, $R_s = 0.042 \mu\text{m}$, $\alpha = 5.7$). There were 2300 events and the bin size was 65 nm. The large sphere was clearly attracted to the vesicle wall, especially where the vesicle was most curved.

electrostatic interactions



$$\Delta G = \Delta H - T\Delta S$$

Enthalpic change

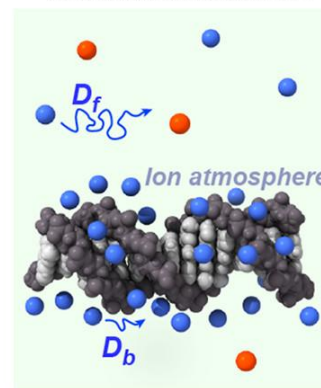
- Electrostatic interactions
 - Hydrogen bonds
- Van der Waals contacts
- Distortion of torsion angles

Entropic change

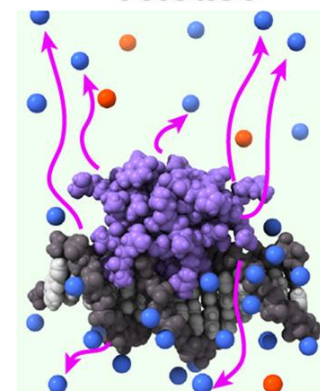
- Hydrophobic effects
- Rotational & translational
- Conformational dynamics
 - Counterion release

$$\frac{P(B)}{P(A)} = \frac{[B]}{[A]} = \exp\left(\frac{-\Delta G}{k_B T}\right)$$

A Ion condensation



B Counterion release



free energy

- Helmholtz:

$$F = E - TS$$

- Gibbs:

$$G = E - TS + PV$$

internal energy

work to create space

thermal energy set by environment

summary last week:

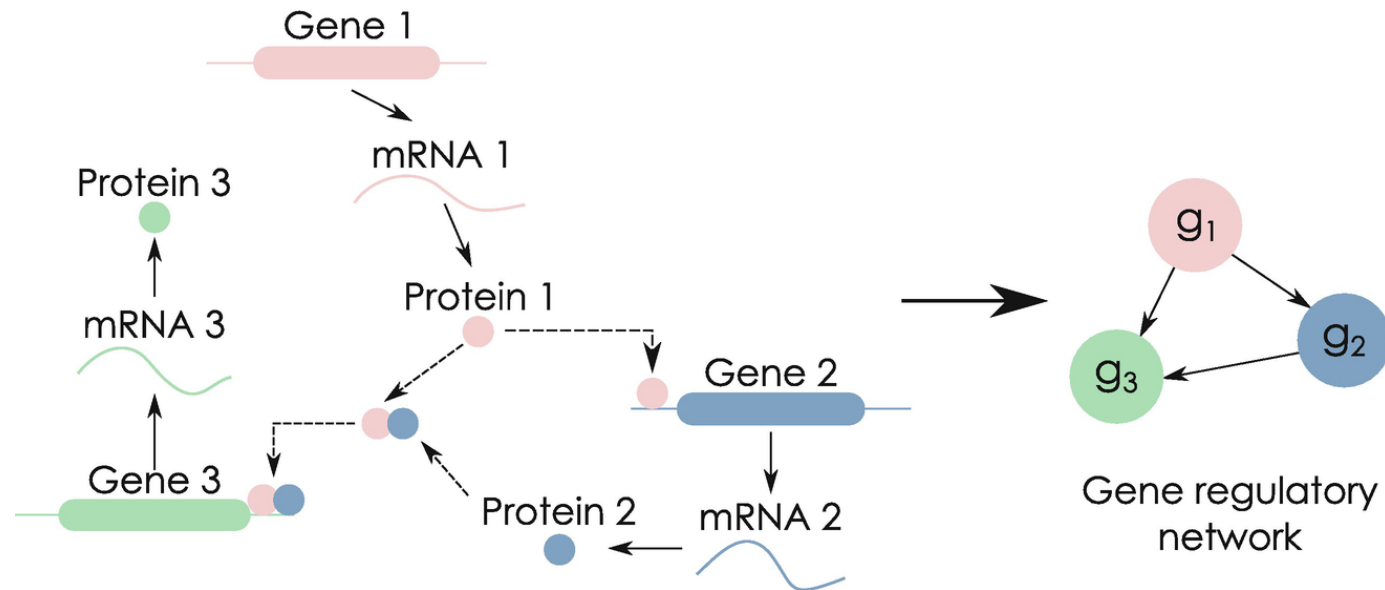
- entropy:
$$S = k_B \ln \Omega = -k_B \sum_{i=1}^N p_i \ln p_i$$
- canonical system: no exchange of particles, but good exchange of energy. In equilibrium: entropy of the outside world couples in by shared temperature
- Question 5D 2f: “The **thermodynamic limit** is the limit in which statistical fluctuations of the system parameters are negligible. What is the minimal number of particles in the system?”

Life's thermodynamic hack: small processes that **appear to decrease entropy** piggyback on simultaneous processes that increase entropy, resulting in a net **increase in disorder**

- Still: single proteins do work in a controlled and directional fashion?
- A single protein will have many conformations?

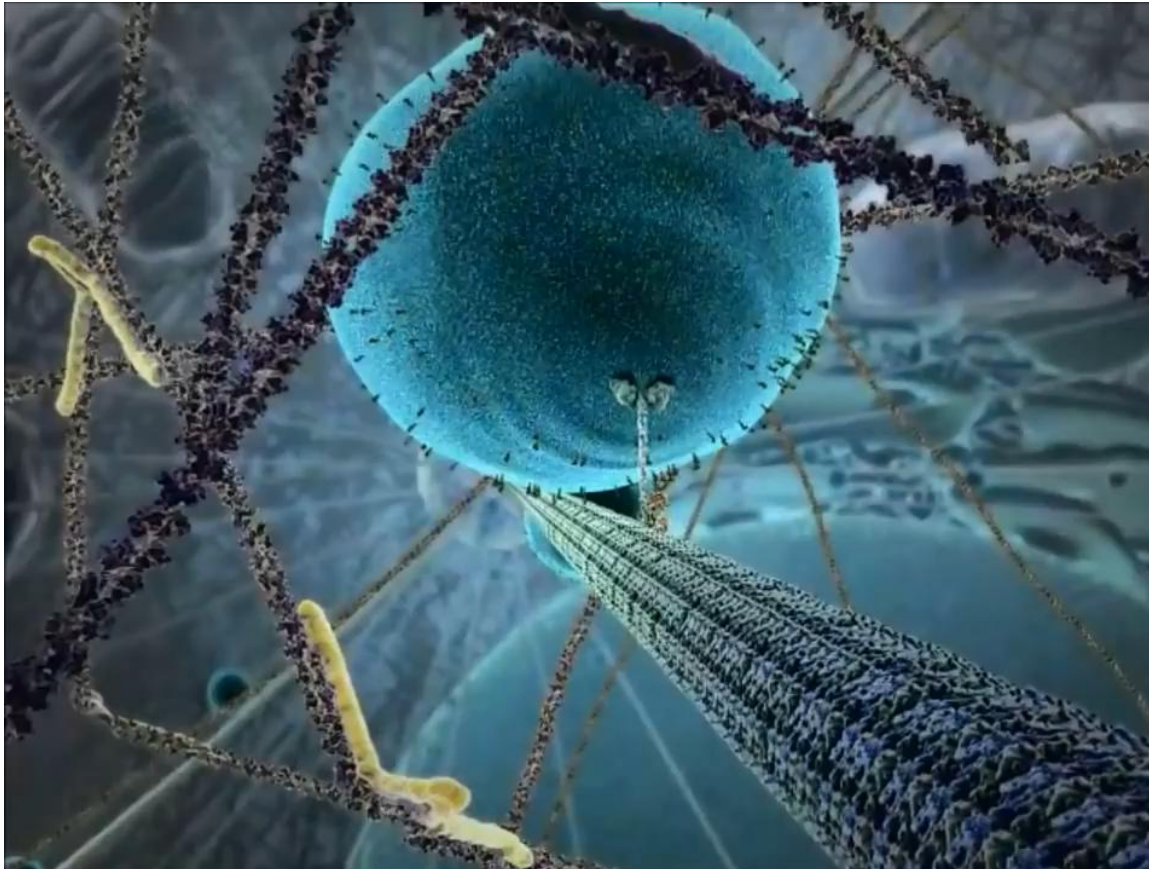
Chapter 6

If all processes yield the highest entropy state, how can a cell control these processes such that they occur in the correct order?



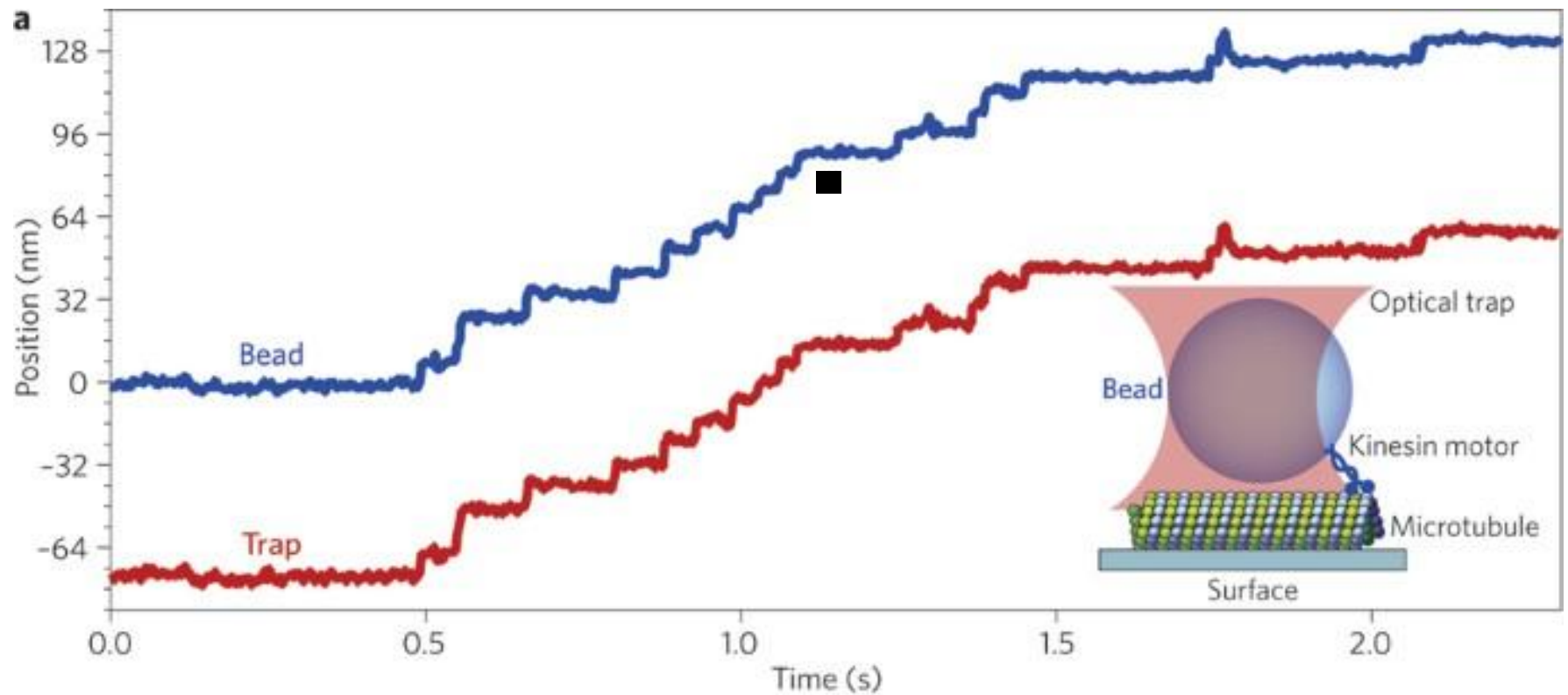
Chapter 4

How can cells distribute their metabolites efficiently despite the fact that friction annihilates directional motion?



<https://www.youtube.com/watch?v=y-uuk4Pr2i8>

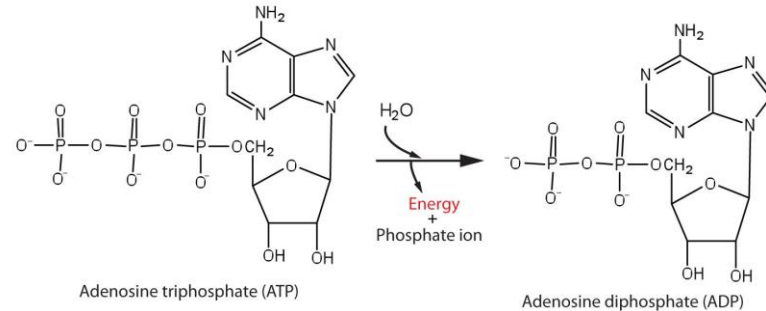
kinesin motors



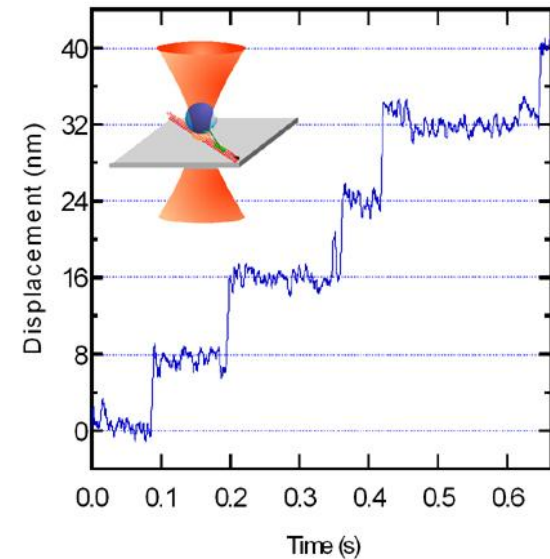
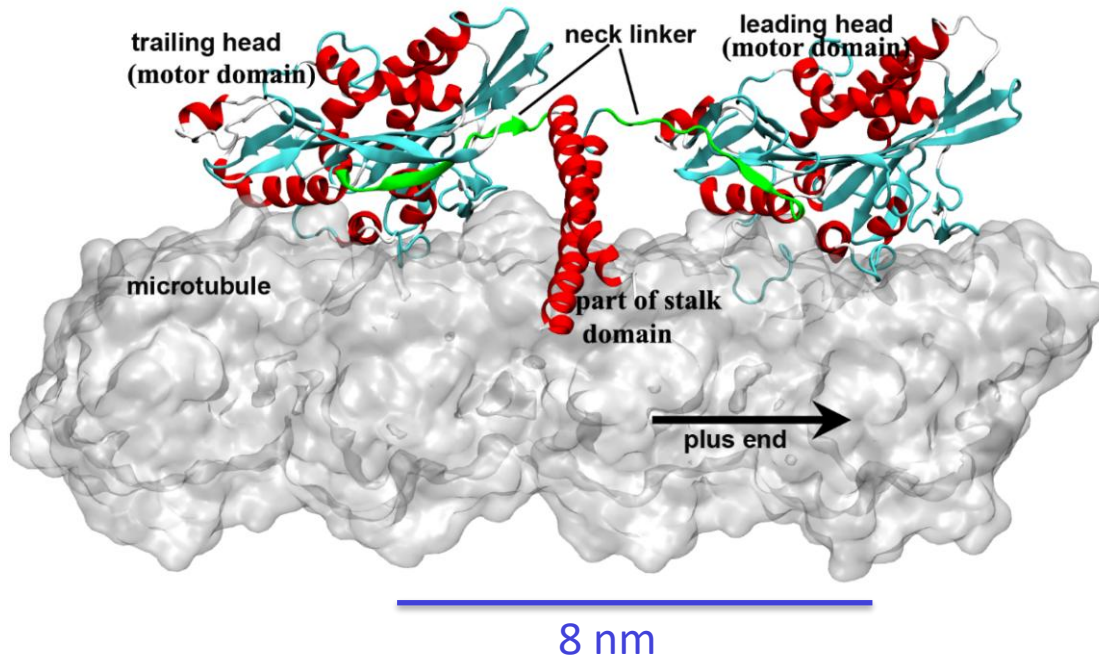
Fazal and Block 2011 Nature Photonics

coupling of two reactions

- chemical transition:

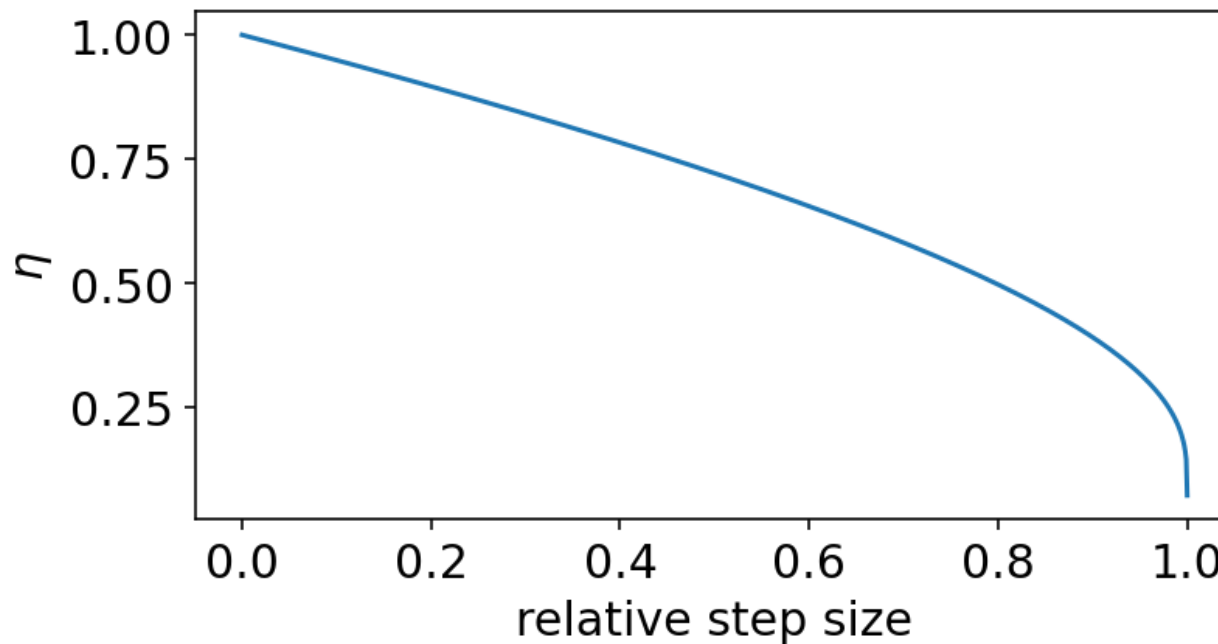


- mechanical transition:



small steps yield high efficiency

- Q 5C 2h:



food to energy

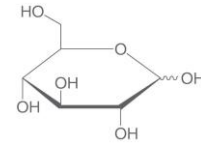


The standard Gibbs free energy (ΔG°) for the complete oxidation of one molecule of glucose to carbon dioxide and water is – **1158 $k_B T$** (= -2,870 kJ/mol).

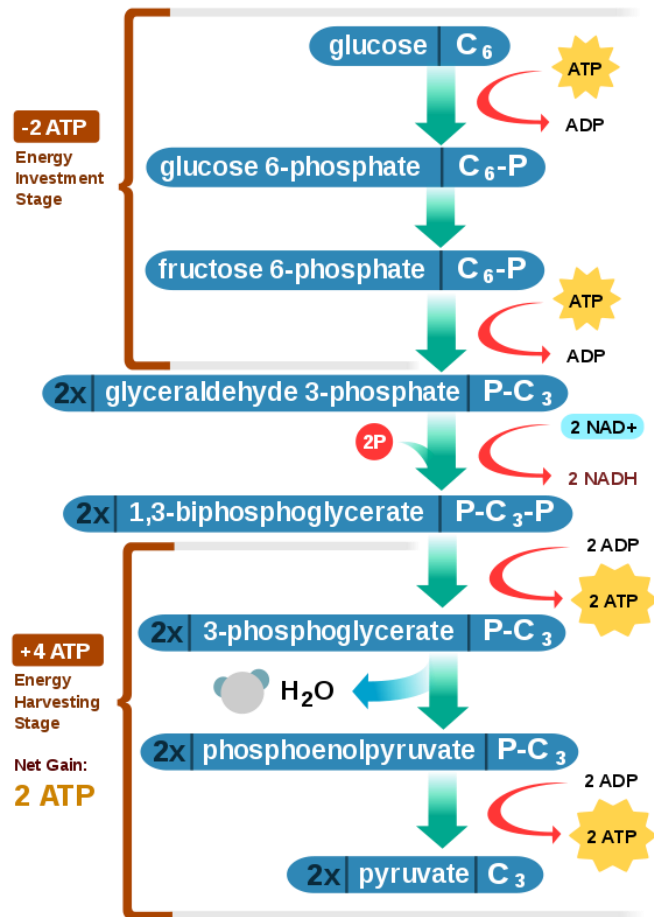
In cellular respiration, about 35 molecules of ATP are produced per molecule of glucose.

Efficiency: 34 - 40%

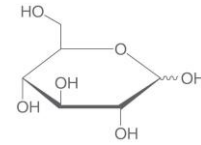
oxidation of glucose



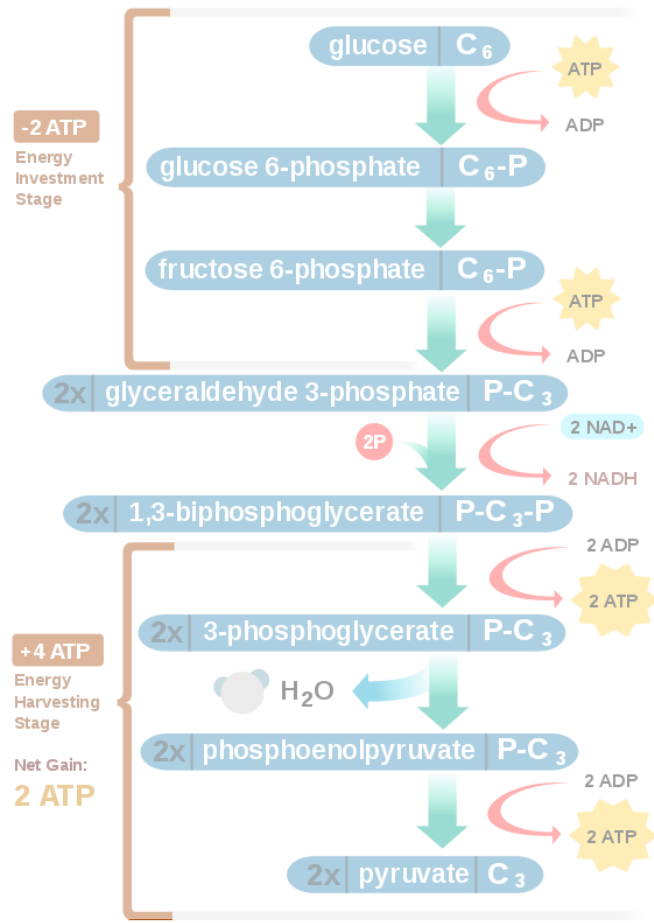
Glycolysis in the Cytoplasm



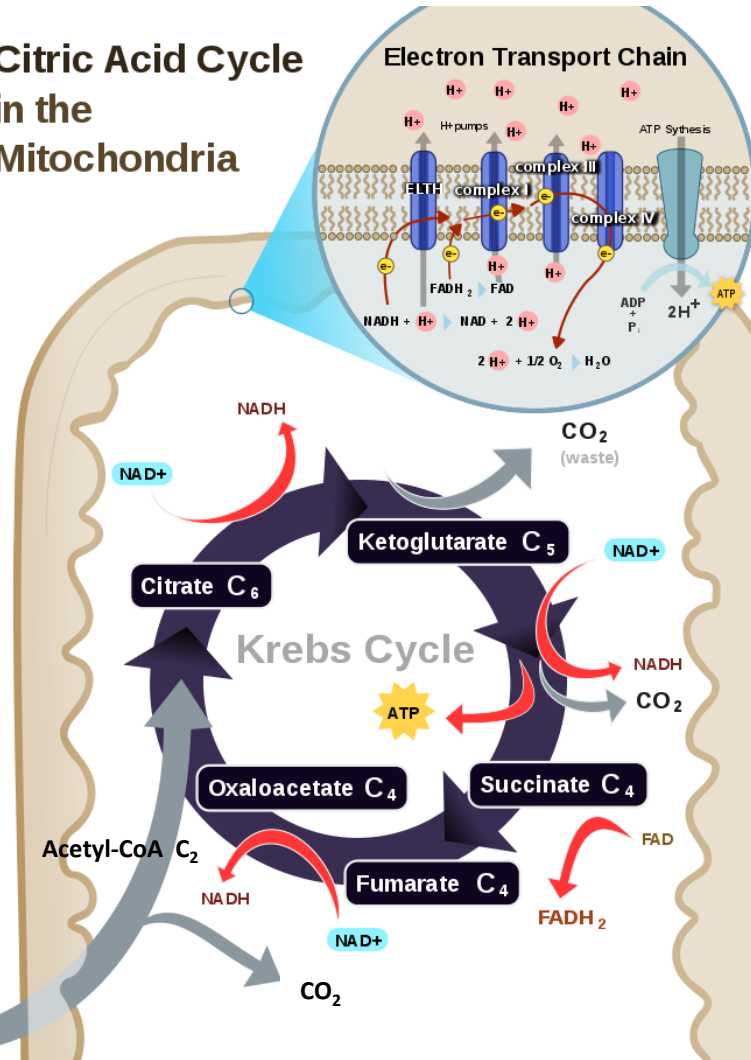
oxidation of glucose



Glycolysis in the Cytoplasm

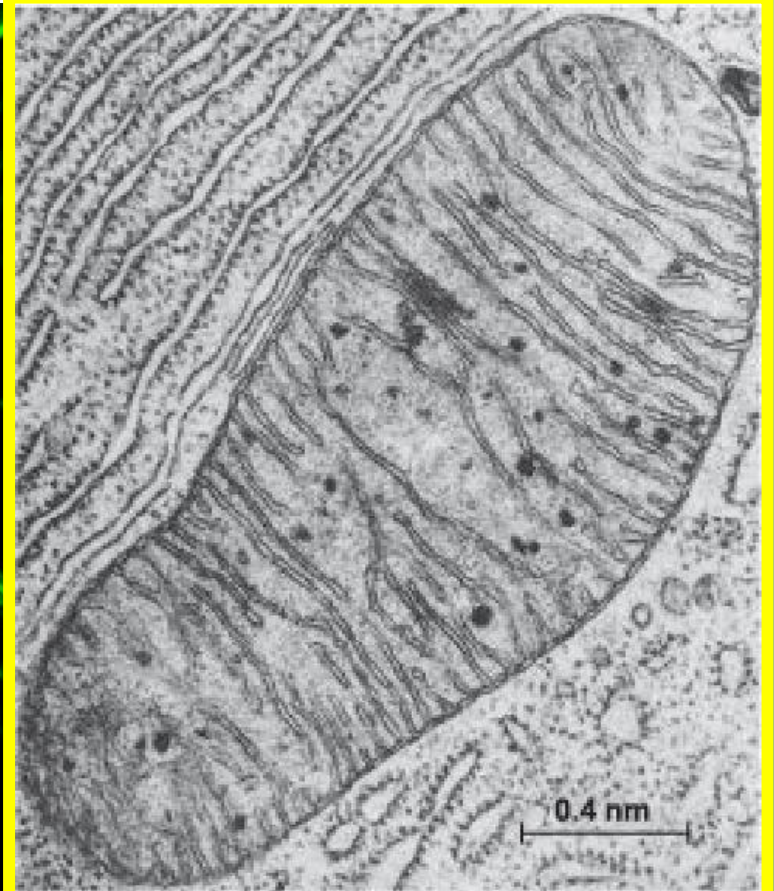
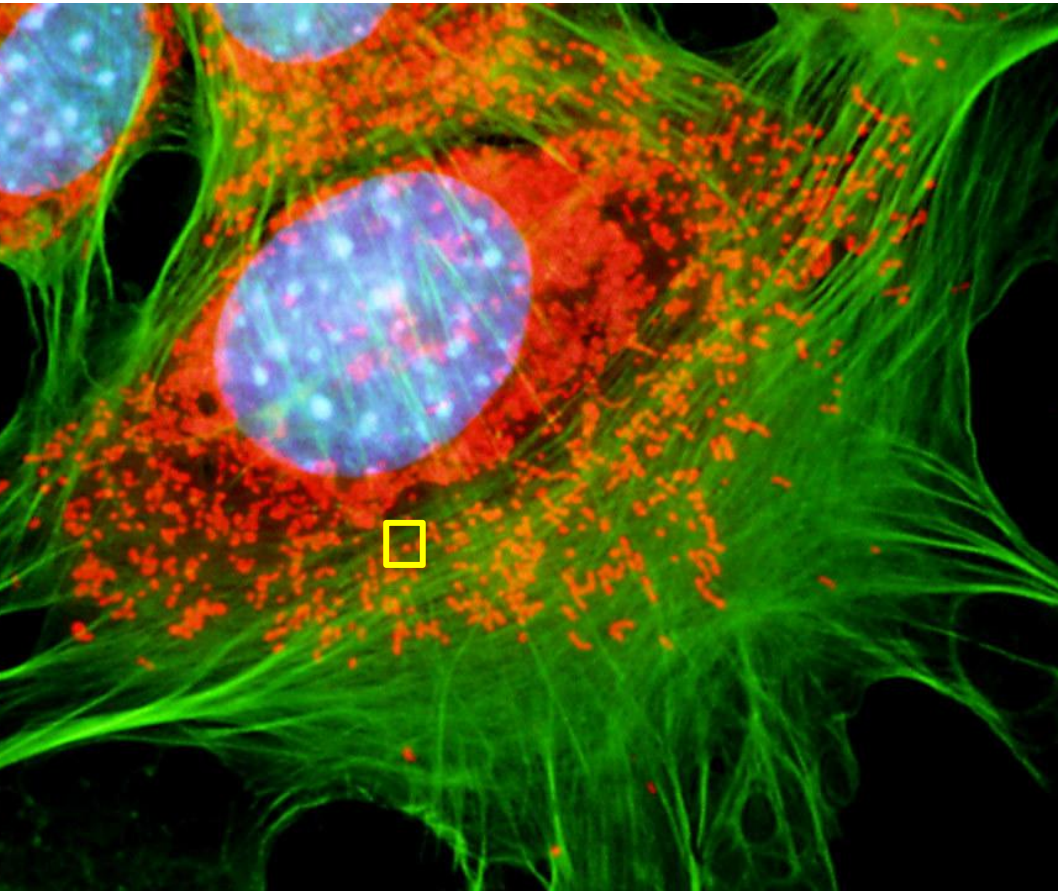


Citric Acid Cycle in the Mitochondria

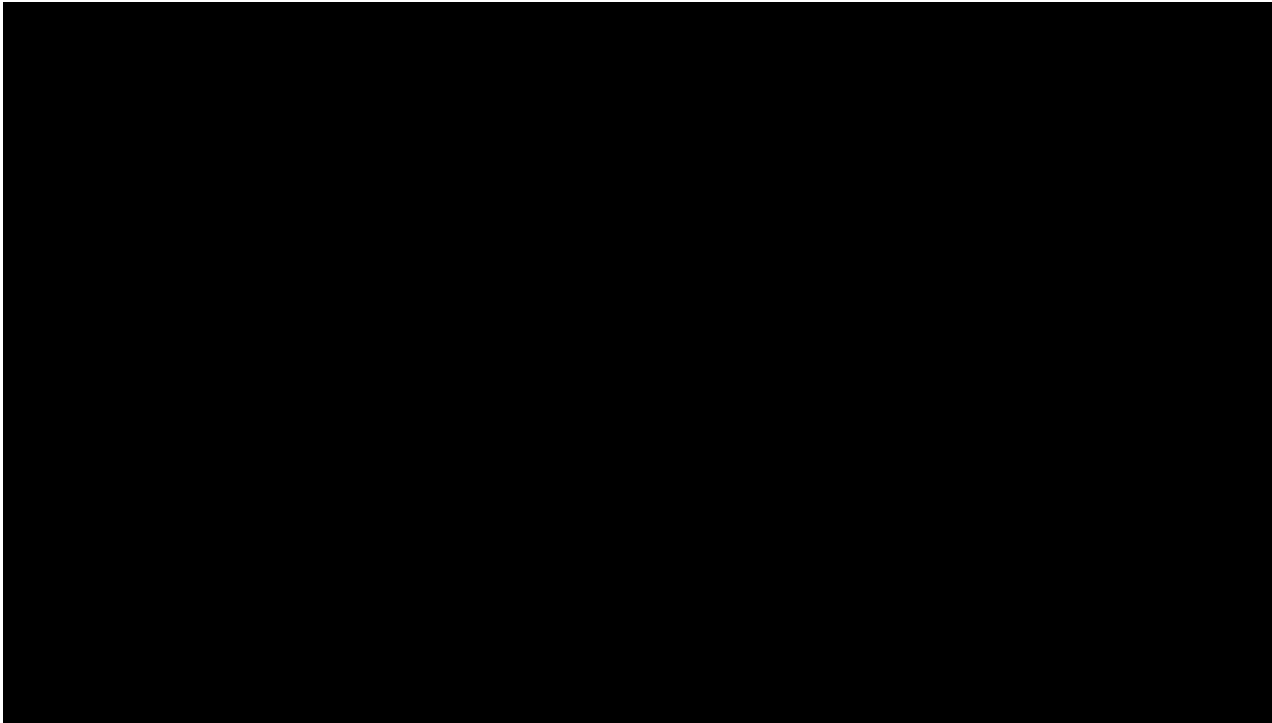


Net yield: ~35 ATP's per glucose molecule

mitochondria produce ATP

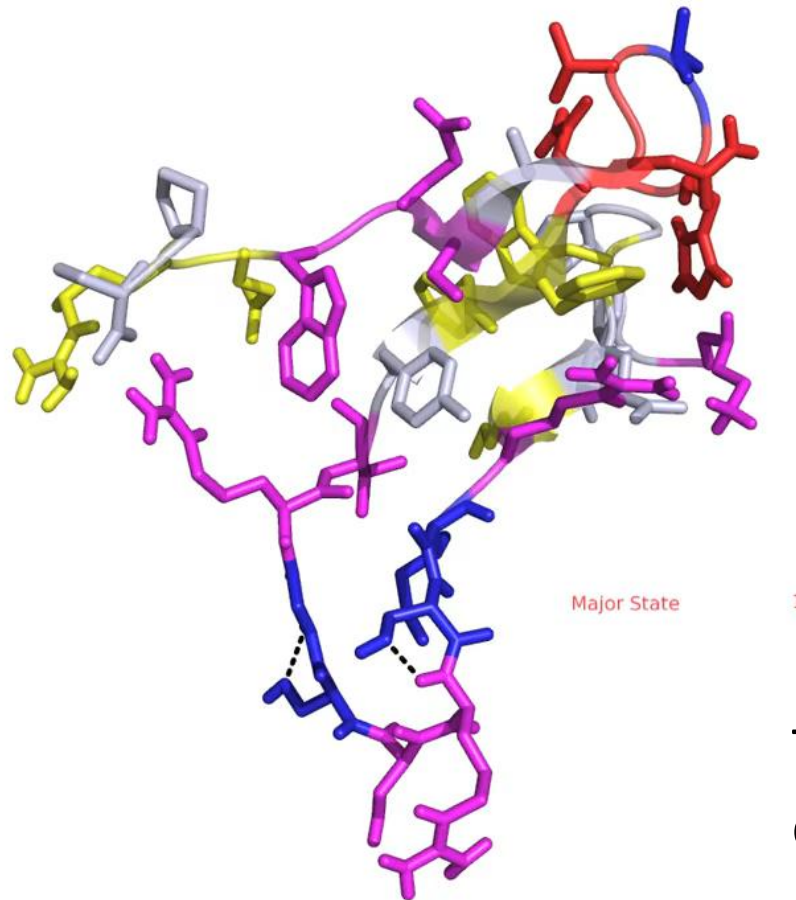


ATP synthase



<https://www.youtube.com/watch?v=kXpzp4RDGJI>

conformational ensembles



Thermal motion
continuously changes
protein conformation

https://en.wikipedia.org/wiki/Conformational_ensembles

average properties

- using 1st moment:

$$\langle E \rangle = \sum_n^N p_i E_i$$

- and Boltzmann's law, with partition function Z

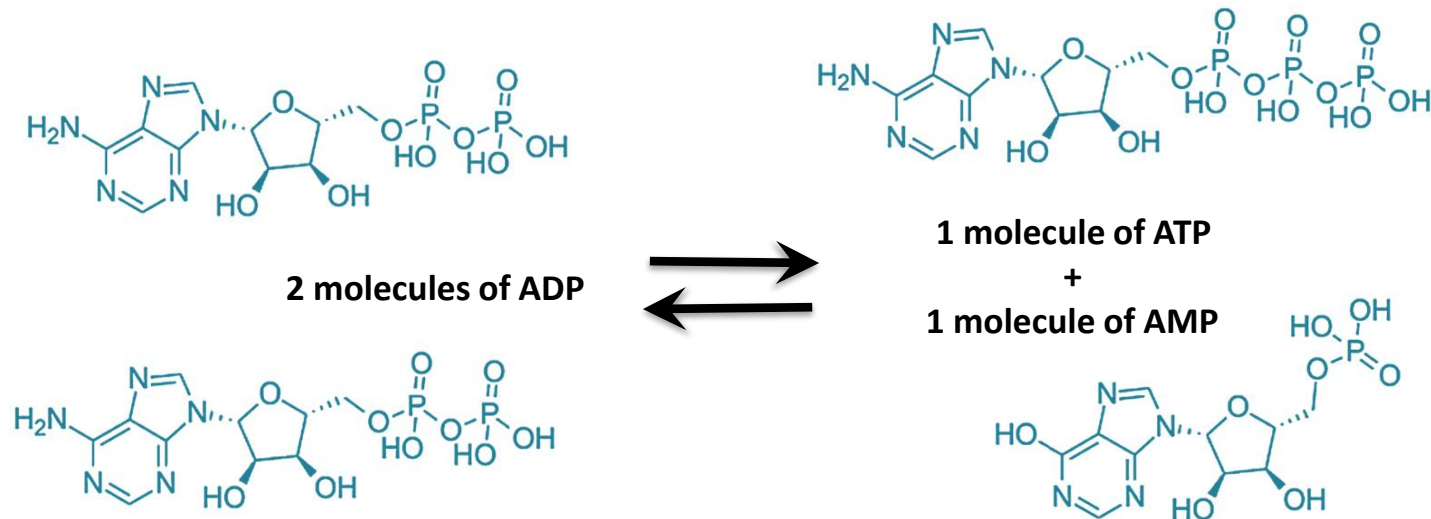
$$\begin{aligned} p_i &= \frac{\exp(-E_i/k_B T)}{\sum_i \exp(-E_i/k_B T)} \\ &= \frac{\exp(-E_i/k_B T)}{Z} \end{aligned}$$

conformational ensembles

- Free energy: $G \equiv \langle E \rangle - TS$

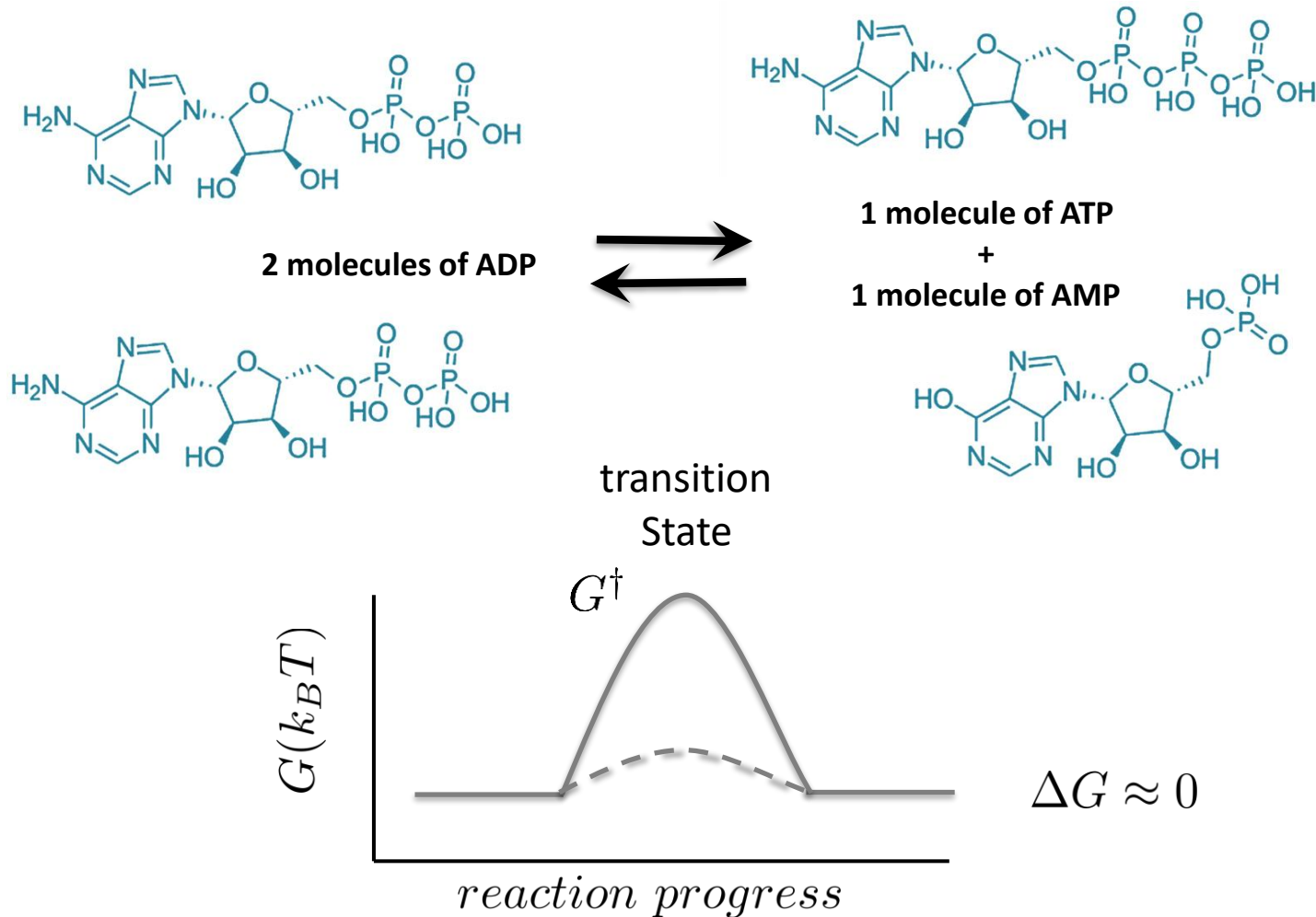
$$\begin{aligned}
 &= \sum_i^N E_i p_i + k_B T \sum_i^N p_i \ln p_i \\
 &= \sum_i^N p_i \{E_i + k_B T \ln p_i\} \\
 &= \sum_i^N \frac{1}{Z} \exp(-E_i/k_B T) \{E_i + k_B T \ln \frac{1}{Z} \exp(-E_i/k_B T)\} \\
 &= \frac{1}{Z} \sum_i^N \exp(-E_i/k_B T) \{E_i - k_B T \ln Z - E_i\} \\
 &= -k_B T \ln Z \frac{1}{Z} \sum_i^N \exp(-E_i/k_B T) \\
 &= -k_B T \ln Z
 \end{aligned}$$

adenylate kinase



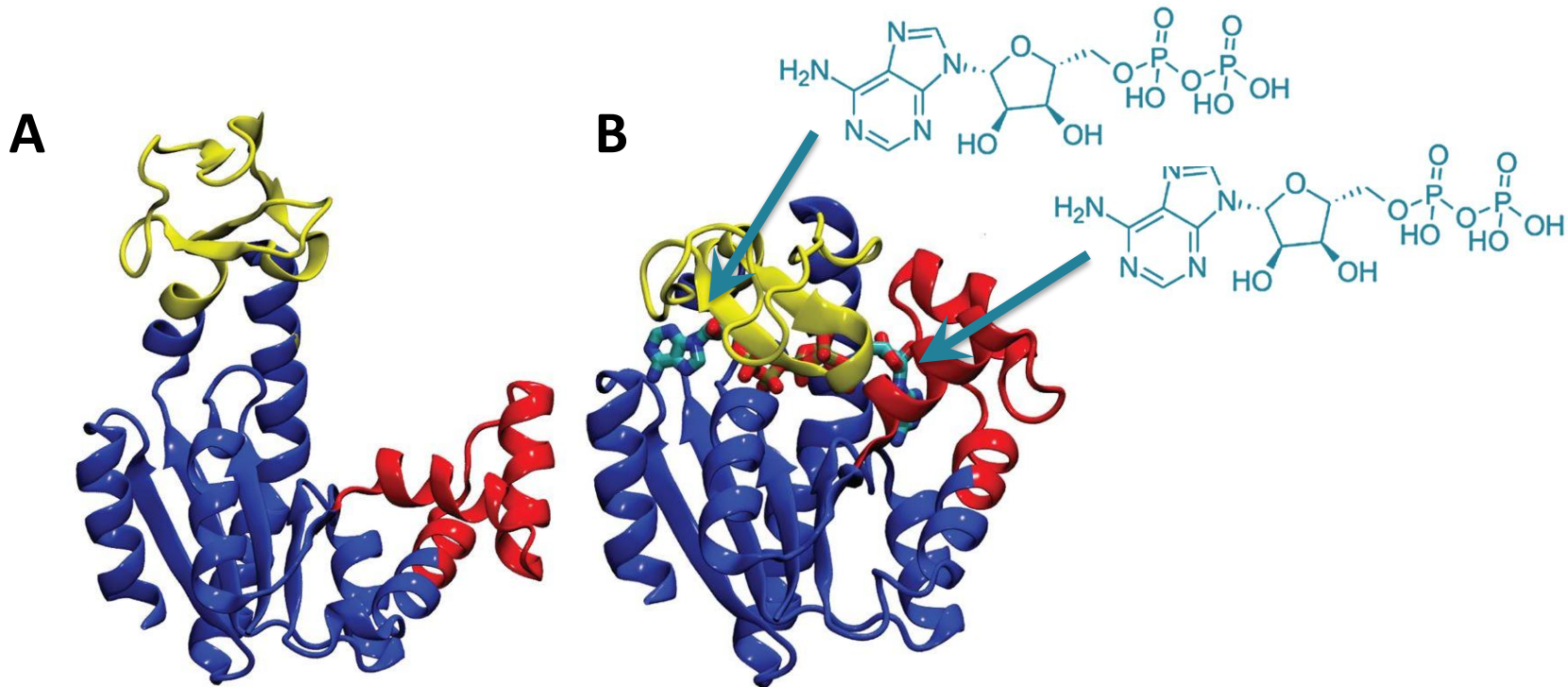
- Catalyses conversion of adenosine phosphates ATP, ADP and AMP:
- Key role in maintaining balance in energy sources
- No change in free energy: $\Delta G \approx 0$

adenylate kinase



- Enzymes reduce energy barrier

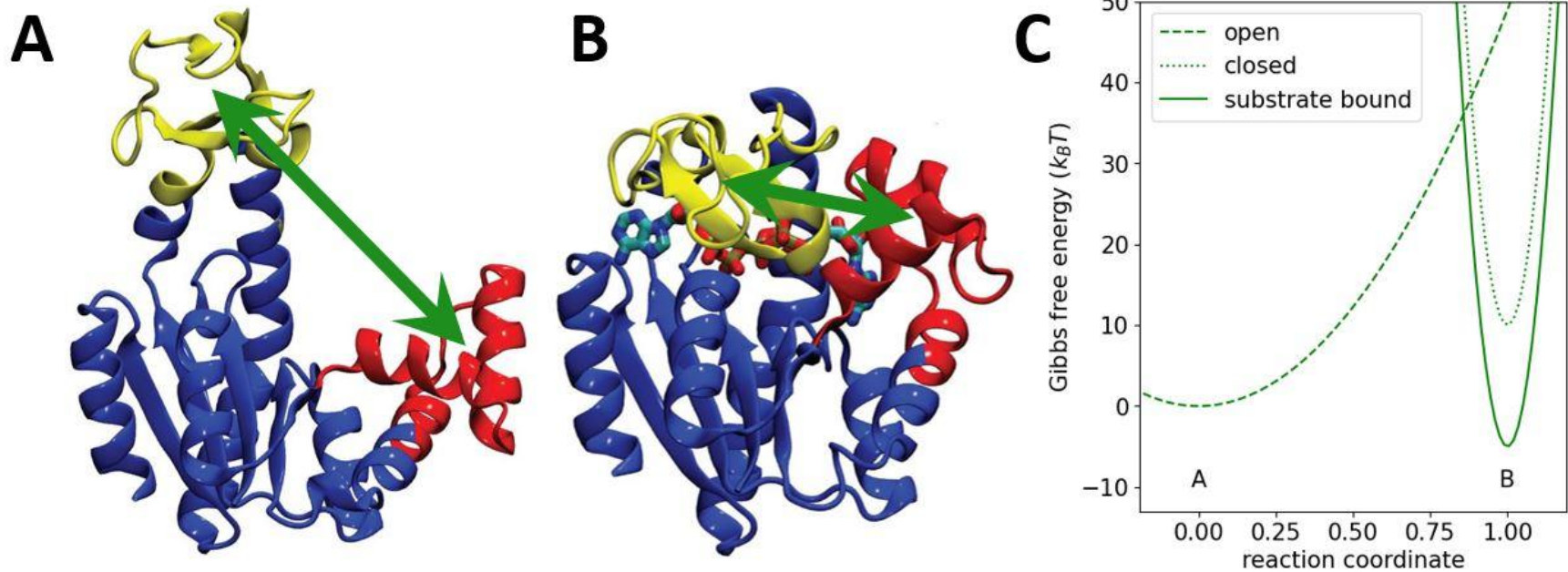
adenylate kinase



Modified from Oradd et al. 2021 Sci. Adv.

2 conformations: with and without substrate

conformational changes



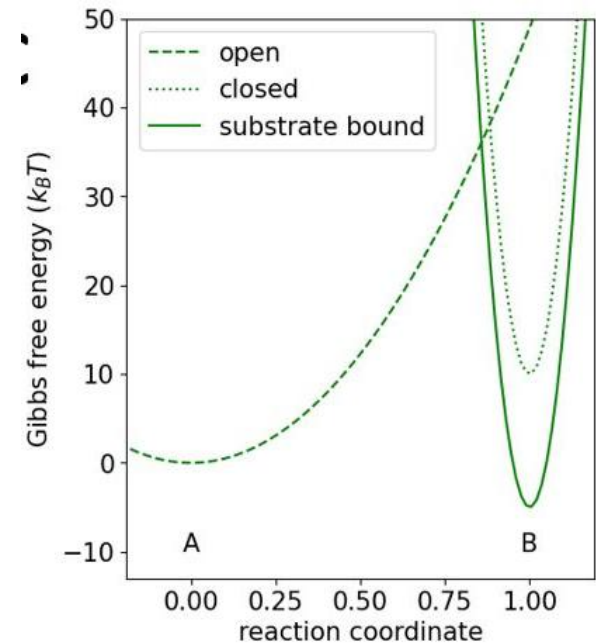
Modified from Oradd et al. 2021 Sci. Adv.

- Reaction coordinate defines the relative displacement of sub-units
- Taylor expansion for conformational changes

conformational changes

- Potential energy:

$$E = \frac{1}{2} \kappa x^2$$



$$\begin{aligned}\kappa_A &= 2E/x^2 \\ &= 2 * 50 * 4.1 \cdot 10^{-21} / (10^{-9})^2 \\ &= 0.41 \text{ N/m}\end{aligned}$$

conformational changes

- Potential energy:

$$E = \frac{1}{2} \kappa x^2$$

- Langevin equation for motion:

$$m \frac{d^2 x}{dt^2} + \xi \frac{dx}{dt} + \kappa x = f(t)$$

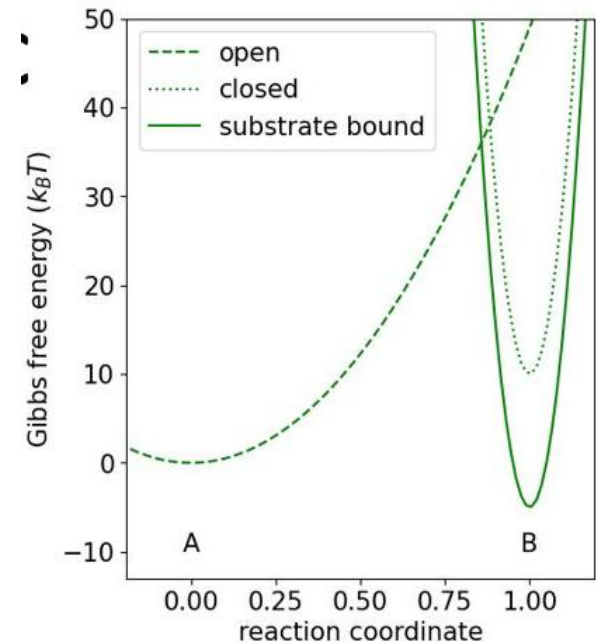
- Solution:

$$x(t) = x(0) \exp\left(-\frac{\kappa}{\xi} t\right)$$

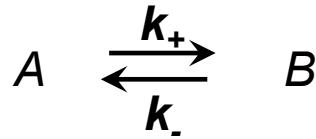
- Time constant:

$$\begin{aligned} \tau_A &= \frac{\xi}{\kappa} = \frac{6\pi\eta r}{\kappa} \\ &= \frac{6\pi \cdot 0.001 \cdot 10^{-9}}{0.4} = 47 \text{ ms} \end{aligned}$$

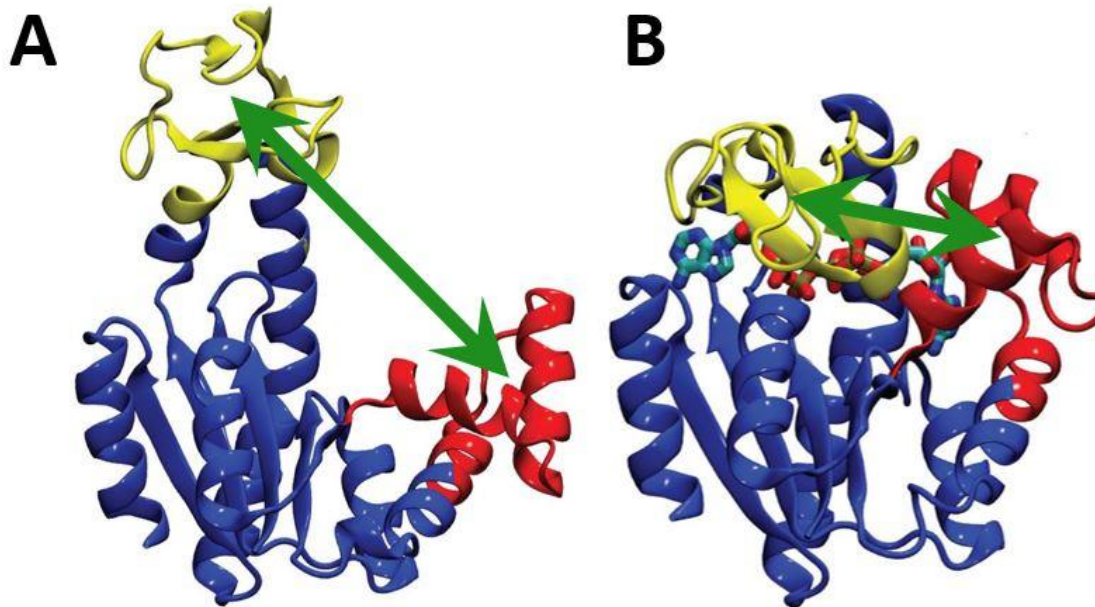
$$\begin{aligned} \kappa_A &= 2E/x^2 \\ &= 2 * 50 * 4.1 \cdot 10^{-21} / (10^{-9})^2 \\ &= 0.41 \text{ N/m} \end{aligned}$$



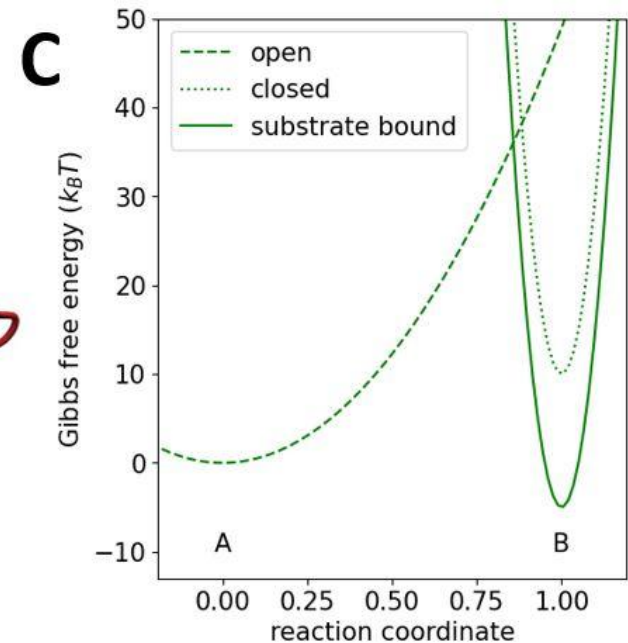
equilibrium



$$\frac{P(B)}{P(A)} = \frac{[B]}{[A]} = \exp\left(\frac{-\Delta G}{k_B T}\right)$$

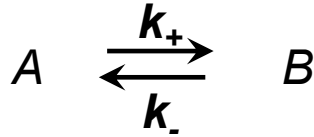


Modified from Oradd et al. 2021 Sci. Adv.



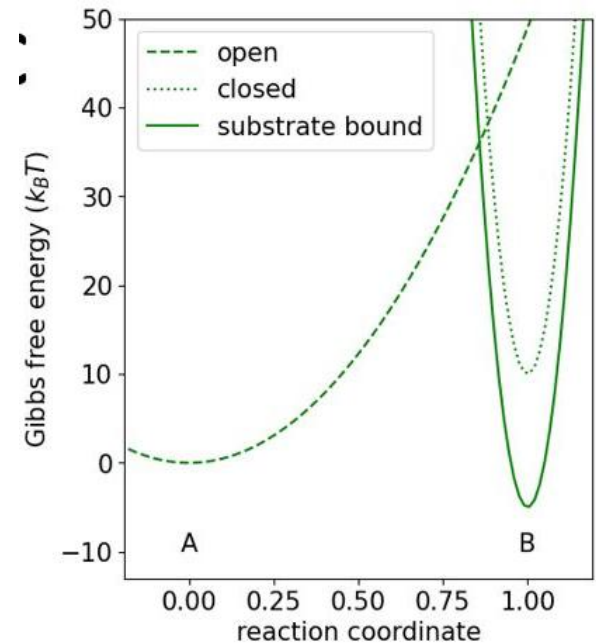
$$\frac{[B]}{[A]} = \exp(-10) = 4.5 \cdot 10^{-5}$$

reaction rates



$$\frac{P(B)}{P(A)} = \frac{[B]}{[A]} = \exp\left(\frac{-\Delta G}{k_B T}\right)$$

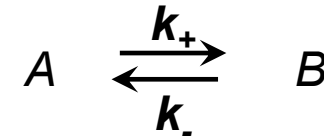
$$\begin{aligned}\frac{k_-}{k_+} &= \exp(-\Delta G_B^\dagger/k_B T) / \exp(-\Delta G_A^\dagger/k_B T) \\ &= \exp(-(\Delta G_B^\dagger - \Delta G_A^\dagger)/k_B T) \\ &= \exp(-(G_B - G_A)/k_B T) \\ &= \exp(-\Delta G/k_B T)\end{aligned}$$



- Ratio of concentrations equals ratio of reaction rates
- Equilibrium constant:

$$K = \frac{k_-}{k_+} = \frac{[B]}{[A]}$$

reaction rates

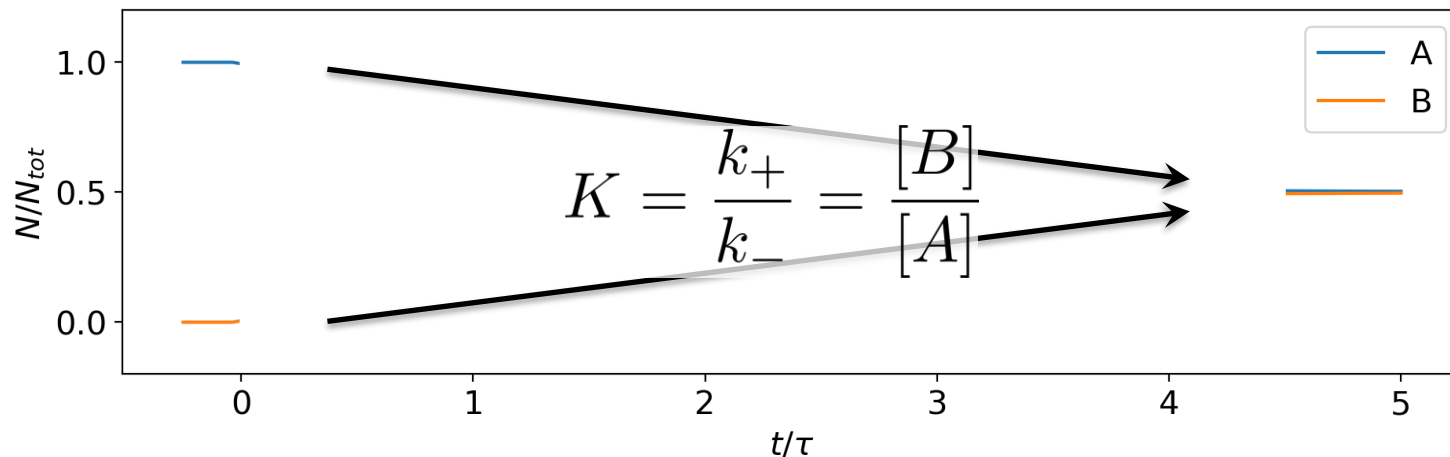


$$\begin{aligned}\frac{dN_A}{dt} &= -k_+ N_A(t) + k_- N_B(t) \\ &= -k_+ N_A(t) + k_- (N_{tot} - N_A(t)) \\ &= -(k_+ + k_-) N_A(t) + k_- N_{tot}\end{aligned}$$

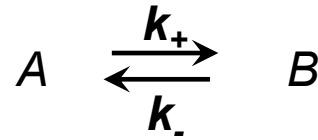
$$\left. \frac{dN_A}{dt} \right|_{eq} = 0$$

$$-(k_+ + k_-)N_{A,eq} + k_- N_{tot} = 0$$

$$N_{A,eq} = \frac{k_- N_{tot}}{(k_+ + k_-)}$$



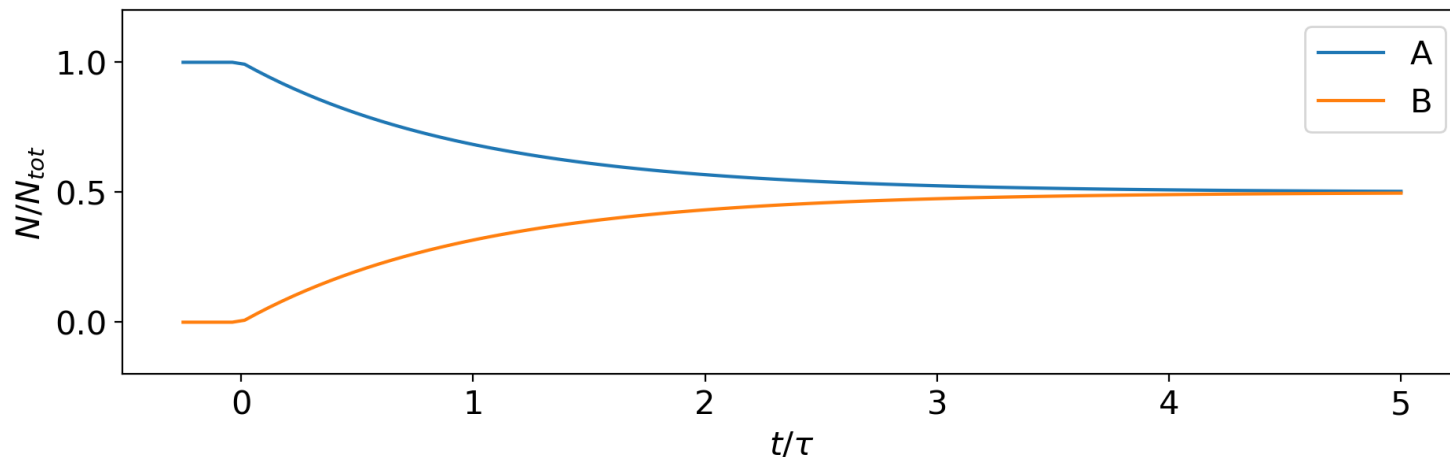
reaction rates



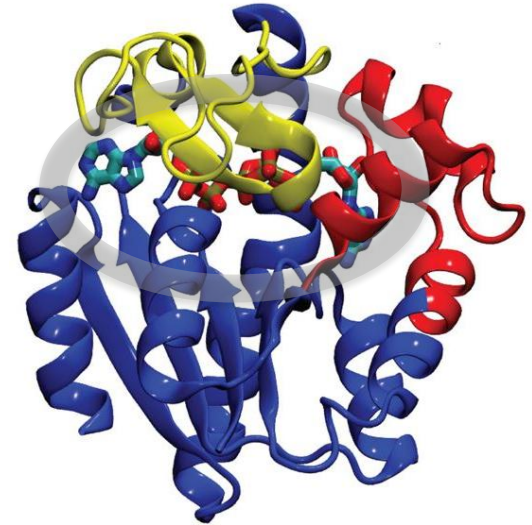
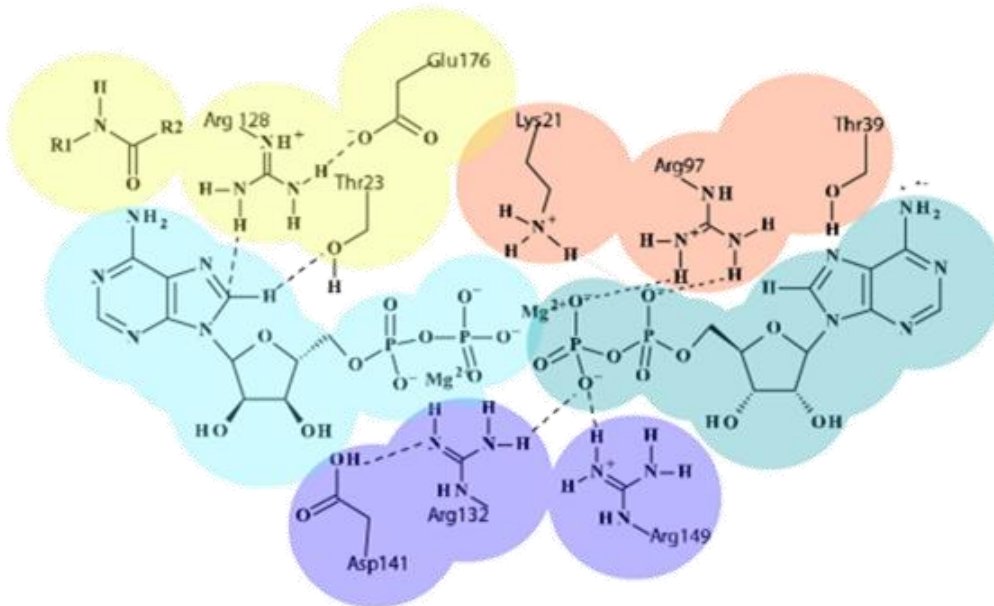
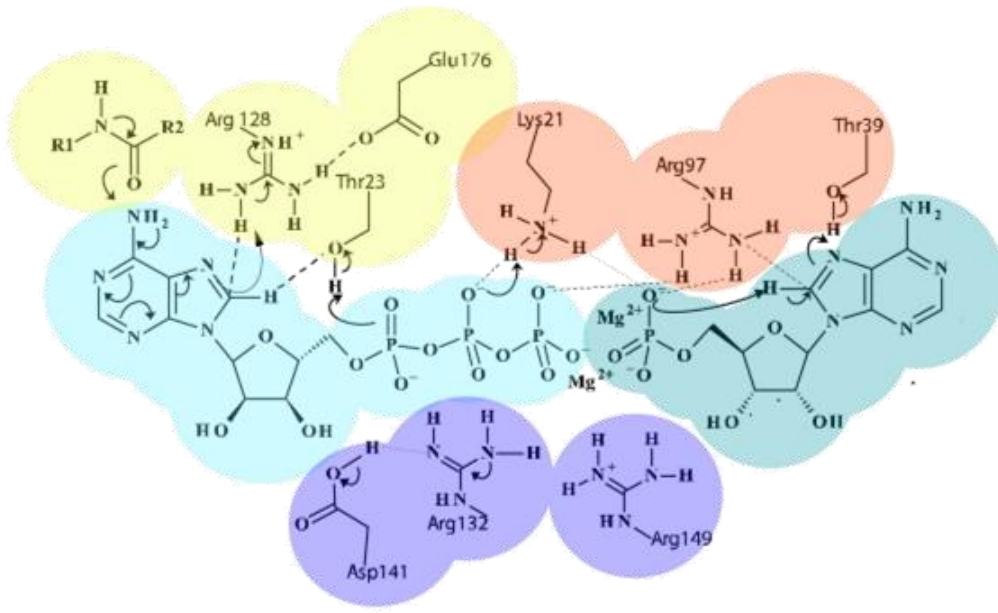
$$\begin{aligned}\frac{dN_A(t) - N_{A,eq}}{dt} &= -(k_+ + k_-) N_A(t) + k_- N_{tot} \\ &= -(k_+ + k_-) N_A(t) + (k_+ + k_-) N_{A,eq} \\ &= -(k_+ + k_-) (N_A(t) - N_{A,eq})\end{aligned}$$

$$N_A(t) - N_{A,eq} = (N_A(0) - N_{A,eq}) e^{-(k_+ + k_-) t}$$

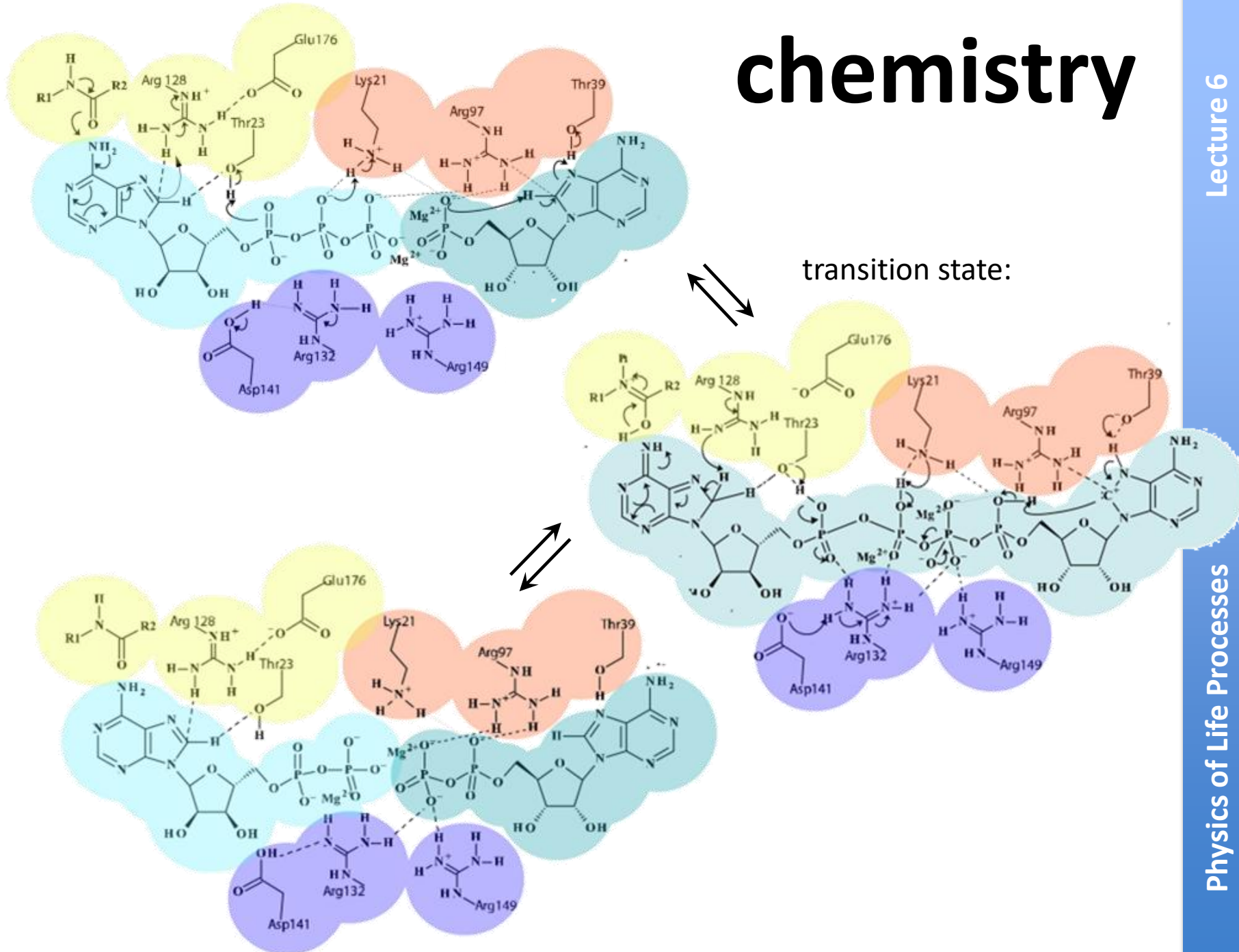
$$N_{A,eq} = \frac{k_- N_{tot}}{(k_+ + k_-)}$$



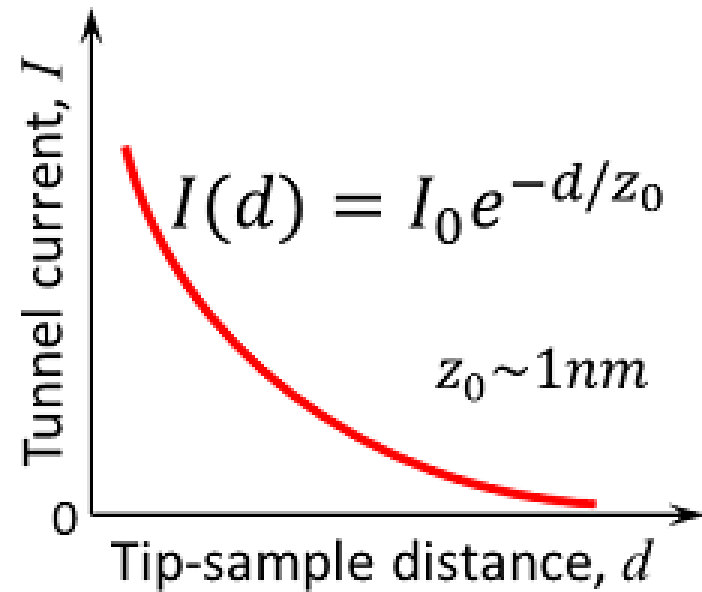
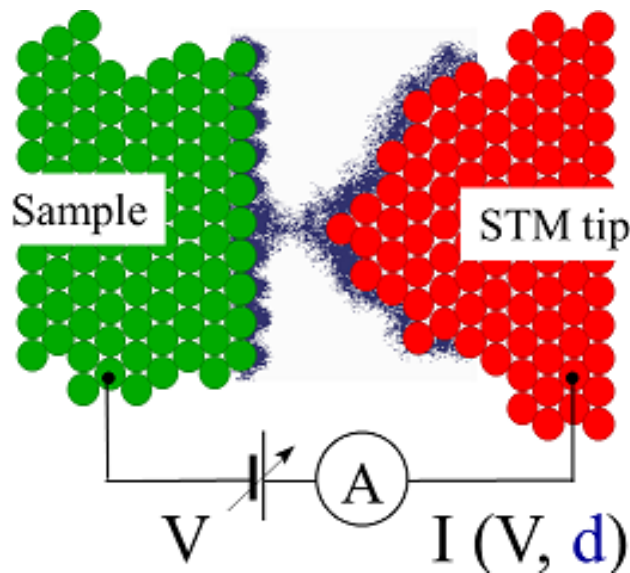
chemistry



chemistry

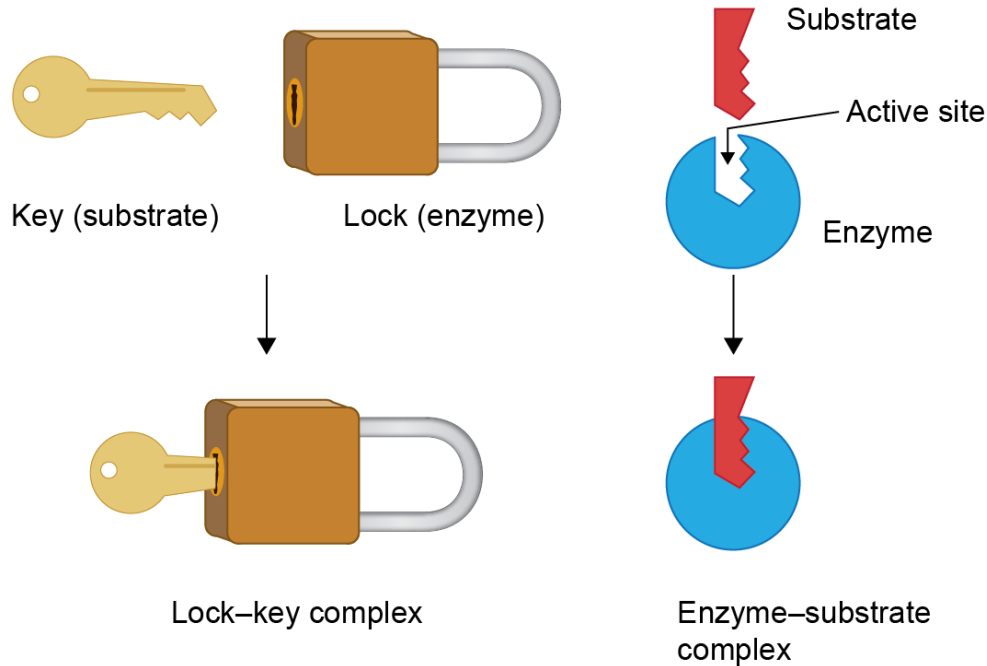


tunneling of electrons



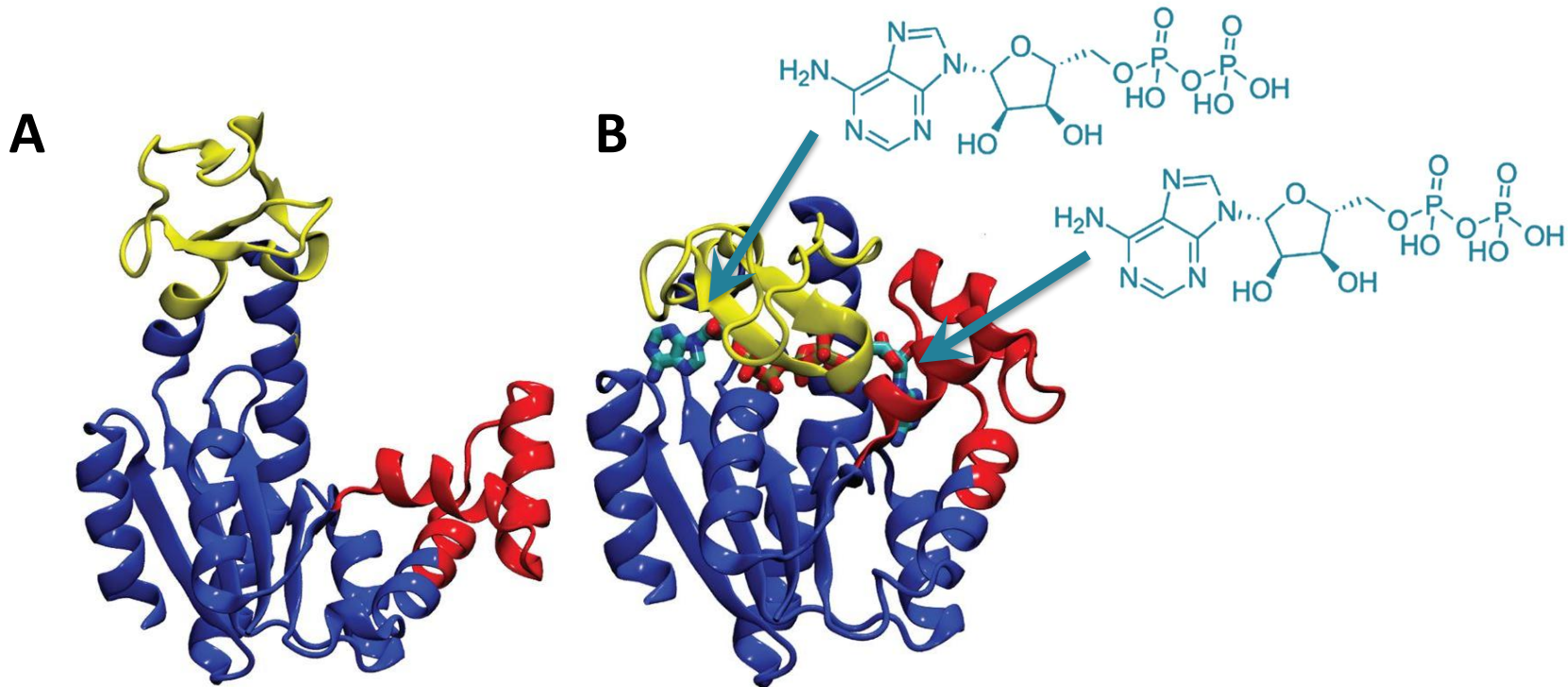
<https://qs.lpem.espci.fr/home/research/underlying-basic-principles/article/stm-sts>

lock-key model



Need for perfect fit!

mechanochemistry



Modified from Oradd et al. 2021 Sci. Adv.

- Chemistry is directly coupled to the mechanics of enzyme
- How about mutations?

question of the week:

- If all processes yield the highest entropy state, how can a cell control these processes such that they occur in the correct order?
- Answer: selective enhancement of reaction rates by enzymes (=proteins encoded in DNA)

De volgende onderwerpen uit de biofysica komen aan bod:

- De inhoud van een cel
- Brownse beweging
- De Stokes-Einstein relatie
- De wet van Fick en de diffusievergelijking
- Entropie en vrije energie
- De Boltzmann-verdeling

Leerdoelen

Na deze cursus kan je kansrekening, statistiek, en combinatoriek inzetten om een variëteit van (bio-)fysische problemen kritisch door te lichten.

Je kunt:

- De programmeertaal Python gebruiken voor berekeningen en simulaties
- Eenvoudige numerieke technieken toepassen op fysische problemen
- Verbanden tussen analytische vergelijkingen en numerieke oplossingen toetsen
- Een beschrijving geven van essentiële onderdelen van een cel
- Statistische parameters afleiden uit “random walks”
- De begrippen entropie en vrije energie uitleggen

Algemene Vaardigheden

Na afloop kan je:

- Constructief commentaar geven op elkaars werk
- Abstracte problemen zelfstandig omzetten in kwantitatieve oplossingen
- Kritisch de nauwkeurigheid van experimentele resultaten evalueren

reflection

further courses

In physics BSc program:

- Complex Physics of Cooking
- Statistical Physics 1

Electives:

- Molecular Biology for Physicists
- Biophysics
- (Physics of Finance)

BSc research project

to do:

- Exercises and test Chapter 6
Deadline: **Friday 15 May 13:00**
- End assignment (Chapter 7)
Deadline: **Friday 22 May 13:00**
- Possible corrections for grades:
Deadline: Monday 1 June 17:00
- Grade submission in USIS:
Deadline: Wednesday 5 June